

AI Solution Design Design Round

AI-Powered Conversational Analytics Assistant for FMCG Beverages

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Executive Summary

This solution proposes an enterprise AI assistant that enables business users to ask natural-language questions about promotional performance, inventory movement, regional sales, and campaign effectiveness. The architecture combines Retrieval-Augmented Generation (RAG), Text-to-SQL, semantic metadata, and a multi-agent workflow to deliver reliable analytical insights while reducing dependency on manual reporting.

Introduction

Business users currently rely on analysts for repetitive reporting, slowing decision-making. The proposed assistant allows users to interact with governed analytical data conversationally and receive KPI summaries, trends, comparisons, and recommendations.

Section A – Problem Decomposition

Supported User Questions	Structured Outputs
Did last month's campaign improve sales in South?	KPI summary with % uplift
Which categories reduced inventory?	Inventory movement table
Compare East vs West sales.	Regional comparison
Top performing products?	Top/Bottom SKU ranking
Promotion ROI?	Insights & recommendations

Section B – AI Solution Approach

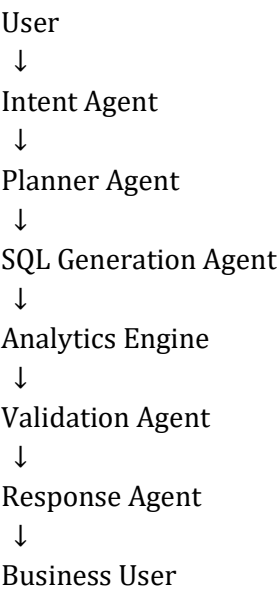
Methodology:

- Large Language Model (LLM)
- Retrieval-Augmented Generation (RAG)
- Semantic Business Layer
- Text-to-SQL Generation
- Multi-Agent Orchestration

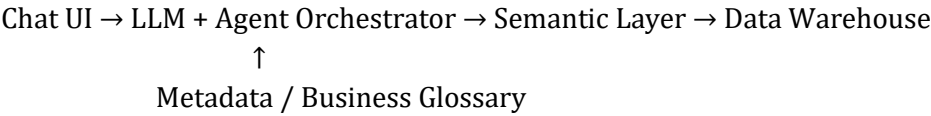
Why suitable:

RAG grounds responses in enterprise data, Text-to-SQL enables analytical querying, and an agentic architecture separates planning, retrieval, execution, and response generation for higher accuracy.

Agentic AI Workflow

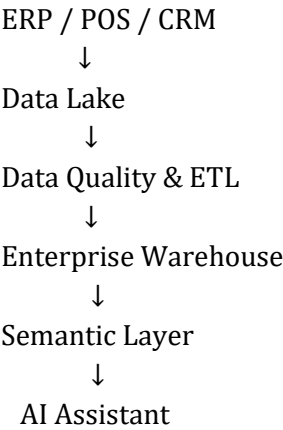


Enterprise Architecture



Section C – Data Interaction Strategy

ETL Workflow



The assistant accesses curated analytical views, generates validated parameterized SQL, executes aggregations in the warehouse, and summarizes the results in business language.

Section D – Limitations & Risks

Risk	Impact	Mitigation
Incorrect query generation	Wrong analytics	Schema-aware prompts, SQL validation
Hallucinated responses	False business conclusions	Ground responses on retrieved data only
Data misinterpretation	Poor decisions	Semantic layer, glossary, confidence checks

Section E – Design Trade-Off

Accuracy over Latency: The design validates SQL, retrieves supporting metadata, and checks outputs before generating the final response. This adds a small delay but significantly improves reliability and trust.

Conclusion

The proposed AI assistant is scalable, secure, and enterprise ready. It empowers business users with conversational analytics, reduces manual reporting effort, and supports faster data-driven decision-making.